



A shock tube and kinetic modeling study of *n*-butanal oxidation

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ABSTRACT

n-Butanal is a key stable intermediate during the combustion of *n*-butanol, and as such strongly affects its chemical kinetics. In this study, ignition delay times of *n*-butanal/oxygen diluted with argon were measured behind reflected shock waves in the temperature range of 1100–1650 K, at pressures of 1.3, 5 and 10 atm, and equivalence ratios of 0.5, 1.0 and 2.0. An *n*-butanal sub-model was developed on the basis of literature review, and exhibits fairly good agreement with the experimental results under all test conditions. Reaction pathway and sensitivity analysis were conducted to gain an insight into the controlling reaction pathways and reaction steps.

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1. Introduction

In recent years, fossil fuel shortage and stringent emission regulations have stimulated investigations of bio-fuels. *n*-Butanol, being a promising bio-fuel or additive, shows relatively good combustion performance and emission-reducing characteristics [1–3]. However, the usage of bio-fuels leads to an increase of toxic compounds such as aldehydes. Some fundamental studies have identified *n*-butanal as a key stable intermediate during the oxidation of *n*-butanol [4,5]. Its production at high temperatures is mainly through decomposition of the radicals that results from the H-abstraction at the α -carbon site or the hydroxyl group. At low temperatures, most of the $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHOH}$ radicals react with oxygen to form *n*-butanal directly.

So far, a number of chemical kinetic models for *n*-butanol [4,6–10] have been developed and validated against various experimental targets. Most of the models include the *n*-butanal chemistry, but there still exists considerable uncertainty. Specifically, the models of Black et al. [7] and Sarathy et al. [9], show poor prediction on the *n*-butanal profiles for the JSR data. The large under-prediction was attributed to the rapid tautomerisation of butenol to butanal that occurs somewhere between the reactor exit and the analytical system. Nevertheless, in order to assess the correctness of this interpretation, their *n*-butanal sub-models need to be reasonably validated. As such, it is necessary to conduct studies on the chemical kinetics of *n*-butanal.

Some experimental and theoretical studies have been conducted on the oxidation of small molecular aldehydes such as formaldehyde [11–14] and acetaldehyde [15–17]. However, only limited studies have been reported on large aldehydes, such as propanal and *n*-butanal. Lifshitz et al. [18] studied the decomposition of propanal behind reflected shock waves over the temperature ranging from 970 K to 1300 K, and proposed a detailed chemical kinetic model which consists of 22 species and 52 elementary reactions. Kasper et al. [19] measured the stable intermediate species and radicals in low-pressure, burner-stabilized, premixed stoichiometric propanal flames. Their observation suggested that the majority of oxygenated intermediates with high molecular weights could be formed by the addition of an alkyl radical to the aldehyde fuel. High-temperature ignition delay times of propanal were measured by Akih-Kumgeh et al. [20] behind reflected shock waves at elevated pressures. A chemical kinetic model developed by the authors yielded fairly good agreement with the experimental results. More recently, Veloo et al. [21] conducted JSR and flame studies on the oxidation of propanal over a wide range of equivalence ratios, temperatures and ambient pressures. A detailed model covering low- and high-temperature kinetics was proposed and validated against JSR, flame speed and ignition data. For the oxidation of *n*-butanal, Davidson et al. [22] measured the only available ignition delay data behind reflected shock waves ($\phi = 0.5$ and 1.0) at pressures of 1.4 and 2.6 atm.

Based on the above considerations, it is worthwhile to conduct specific experiments on the oxidation of *n*-butanal and optimize the corresponding sub-model. In this study, ignition delay times of *n*-butanal were measured behind reflected shock waves over a wide range of equivalence ratios, temperatures and pressures.

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A detailed *n*-butanal sub-model was assembled on the basis of literature review and then validated against the experimental results.

2. Experimental approach

The measurements in this study were carried out in a shock tube with 11.5 cm inner diameter. It is separated into a 4.0 m long driver section and a 4.8 m long driven section by double PET (polyester terephthalate) diaphragms. Extensive details about the apparatus and technique are available in Ref. [10]. Fuel mixtures are prepared manometrically in a 128 L stainless steel tank and allowed to mix for at least 12 h by molecular diffusion. To minimize the possibility of fuel condensation, the partial pressure of *n*-butanal is below 50% of its saturation vapor pressures (11.54 kPa) at room temperature. The purities of liquid *n*-butanal, argon, oxygen and helium are 99.5%, 99.995%, 99.995% and 99.999%, respectively. The ignition delay time is defined as the time interval between the arrival of the incident shock wave at the endwall and the extrapolation of the steepest rise in the endwall OH* chemiluminescence signal to the zero baseline, as shown in Fig. 1. The incident shock velocity at the endwall is determined by linear extrapolation of three time intervals recorded by three time counters (Fluke, PM6690). The time counters are triggered by four pressure transducers (PCB, B26) located at the side-wall of the shock tube. The OH* chemiluminescence selected by a narrow filter centered at 306 ± 10 nm is measured with a photomultiplier (Hamamatsu, CR131) located at the endwall. All data are recorded using a digital recorder (Yokogawa, scopecorder-DL750). The temperature behind

the reflected shock wave is calculated by using the reflected shock module in the software Gaseq [23]. Uncertainty of the temperatures is about ± 25 K. The calculation of ignition delay times was performed using Chemkin II [24]. Similar to the experiment, the calculated ignition delay time was defined as the time interval between the onset of reaction and the extrapolation of the steepest rise in the calculated OH* profile to the zero baseline. The effect of pressure-rise was not considered in the simulation due to the relatively short ignition delay times (less than 1.5 ms) and high dilution with argon.

n-Butanal experiments were performed at equivalence ratios of 0.5, 1.0 and 2, pressures of 1.3–10 atm, and temperatures of 1100–1650 K. The composition of the fuel/oxygen/argon mixture is given in Table 1. Most of mixtures are prepared at a dilution level of $\sim 95\%$ except for one series which has a dilution of 85%.

3. Kinetic modeling

The improved *n*-butanal sub-model was developed based on the C4 chemistry [25] proposed by the Curran group, noting that the original *n*-butanal model is relatively rough and lacks some critical reactions. In the sub-model, reactions were added or modified based on literature values.

Bond dissociation energies (BDEs) of *n*-butanal, as well as form-aldehyde, acetaldehyde and propanal, were calculated by da Silva and Bozzelli [26] with CBS-APNO, G3 and G3B3 theoretical method, as shown in Fig. 2. It was found that the R-CH₂CHO bond is the weakest bond in all aldehydes larger than acetaldehyde. A comparison of the BDEs of these aldehydes yields useful information for the kinetic modeling of *n*-butanal. It is observed that the length of carbon chain exerts small effect on the BDEs (~ 89 kcal/mol) of C–H bonds in the formyl group of aldehydes. The same phenomenon also occurs in the R-CHO bonds for aldehydes larger than acetaldehyde, with the BDEs being about 84 kcal/mol. For aldehydes larger than propanal, the formyl group exerts little influence on the terminal primary carbon (γ -carbon) which has very similar C–H bond BDEs (~ 102 kcal/mol) with those of the corresponding alkanes. In addition, the C–H bond of *n*-butanal at the secondary carbon site (β -carbon) also exhibits a similar BDE (~ 99 kcal/mol) with that of alkanes. Therefore, an analogy method can be used in the construction of the *n*-butanal mechanism before the

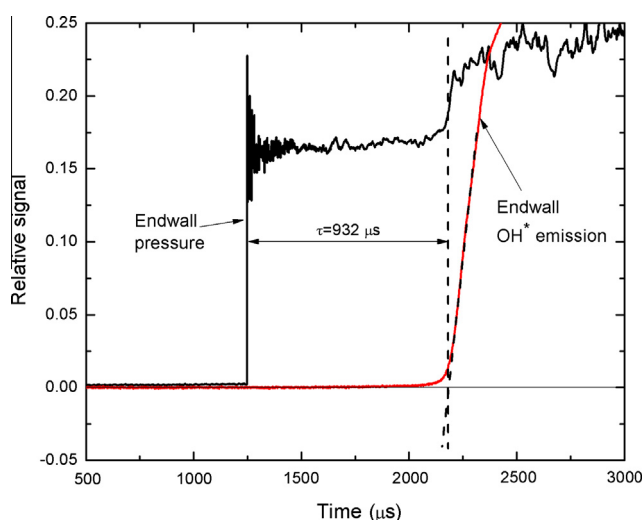


Fig. 1. Typical endwall pressure and OH* chemiluminescence measurements with corresponding ignition delay time for stoichiometric 0.6% *n*-butanal at 5.16 atm and 1246 K.

Table 1
Mixture compositions in this study.

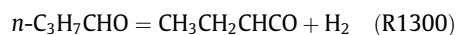
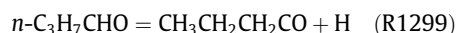
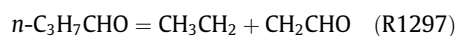
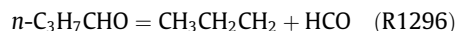
<i>n</i> -Butanal (%)	O ₂ (%)	ϕ	p_5 /atm
1.2	3.3	0.5	1.3
1.2	6.6	1.0	1.3
1.2	13.2	2.0	1.3
0.6	1.65	0.5	5
0.6	3.3	1.0	5
0.6	6.6	2.0	5
0.6	1.65	0.5	10
0.6	3.3	1.0	10
0.6	6.6	2.0	10

Formaldehyde	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C} \\ \\ \text{O} \end{array}$
Acetaldehyde	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C} \\ \quad \\ \text{H} \quad \text{O} \end{array}$
<i>n</i> -Propanal	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{O} \end{array}$
<i>n</i> -Butanal	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{O} \end{array}$

Fig. 2. Bond dissociation energies (BDEs) for C–H and C–C bonds in the aldehydes [26].

availability of high-level calculation of the rate constants of the unimolecular decomposition and H-abstraction reactions.

In this study, the following unimolecular decomposition reactions were considered in the model:



The rate constants for reactions R1296–1299 were calculated from microscopic reversibility using an estimate of the rate constant for radical–radical recombination. This method was also employed by Johnson et al. [27] and Black et al. [7] for the decomposition of propanol and *n*-butanol, respectively. For reaction R1296, a rate constant of $1.204 \times 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used as recommended by Tsang [28] for formyl radical recombination with *n*-propyl radical. Reaction R1297 was estimated to be $2.0 \times 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ as recommended by Tsang [28] for the recombination of ethyl and *n*-propyl radicals to form *n*-pentane. It is noted that the rate constants for reactions R1296 and R1297 have been respectively increased by 50% and 30% to improve the modeling results for ignition delay times, which are not beyond their respective uncertainty limits. For reaction R1298, a rate constant of $1.93 \times 10^{14} \times T^{-0.32} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ recommended by Tsang [28] for the recombination of methyl and *n*-propyl radicals to form *n*-butane was used. The rate constant for reaction R1299 was estimated to be $1.0 \times 10^{14} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ as recommended by Johnson et al. [27] and Black et al. [7] for hydrogen atom addition to other free radicals. The rate constant of reaction R1300 was obtained from an analogy with that of acetaldehyde conducted by Yasunaga et al. [16]. Finally, it is noted that pressure dependent treatment was not performed for these unimolecular decomposition reactions in this study. Extended RRKM treatment is expected to improve the performance of the model in a wider range of pressures.

As mentioned above, the analogy method can be used for the assignment of rate constants of H-abstraction reactions. In this study, rate constants for the H-abstraction from the formyl group and its neighboring carbon site were obtained on analogy with the counterparts of acetaldehyde. The rate constants for H-abstraction from the primary carbon and its neighboring carbon sites of propane included in the C4 model [25] were employed for those of *n*-butanal. Nevertheless, there is a particular reaction that has quite different rate constants from different authors. Specifically, the rate constant of reaction R1300 ($n\text{-C}_3\text{H}_7\text{CHO} + \text{H} = \text{CH}_3\text{CH}_2\text{CH}_2\text{CO} + \text{H}_2$) can be obtained on analogy with either acetaldehyde: $2.37 \times 10^{13} \times \exp(-3.642 \text{ kcal mol}^{-1}/RT) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ by Gupte et al. [29], $1.31 \times 10^5 \times T^{2.58} \exp(-1.22 \text{ kcal mol}^{-1}/RT) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ by Sivaramakrishnan et al. [30] and $3.97 \times 10^6 \times T^{2.15} \exp(-1.59 \text{ kcal mol}^{-1}/RT) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ by Bentz et al. [17], or propanal: $5.54 \times 10^{12} \times T^{3.5} \exp(-5.167 \text{ kcal mol}^{-1}/RT)$ by Lifshitz et al. [18]. As shown in Fig. 2, the C–H bond in the formyl group has the weakest BDE among the C–H bonds. Consequently, it is expected that the H-abstraction from this site plays a dominant role in fuel consumption and have relatively higher reaction rate than those from other sites. A similar analysis was also conducted in the kinetic modeling of propanal by Veloo et al. [21]. Recognizing that the rate constant of H-abstraction from the primary carbon is more reliable, the abstraction from the formyl group should have relatively higher reaction rate and lower activation energy. Figure 3 shows the reaction rate constants for reaction R1300 from different authors, as well as the H-abstraction rate constants from the

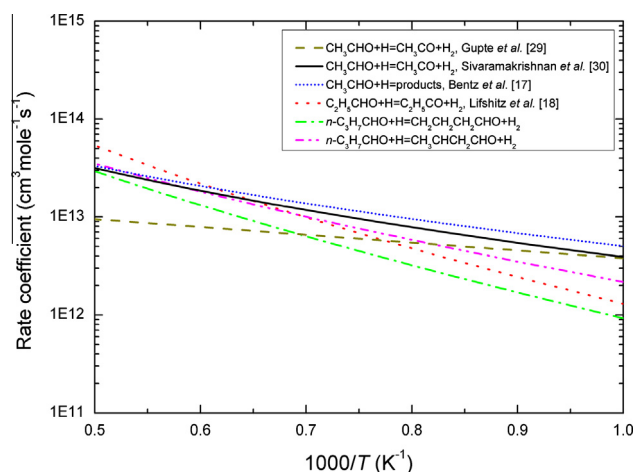


Fig. 3. Reaction rate constants for $n\text{-C}_3\text{H}_7\text{CHO} + \text{H} = \text{CH}_3\text{CH}_2\text{CH}_2\text{CO} + \text{H}_2$ from different authors.

primary (R1302: $n\text{-C}_3\text{H}_7\text{CHO} + \text{H} = \text{CH}_2\text{CH}_2\text{CH}_2\text{CHO} + \text{H}_2$) and β carbon sites (R1303: $n\text{-C}_3\text{H}_7\text{CHO} + \text{H} = \text{CH}_3\text{CHCH}_2\text{CHO} + \text{H}_2$). The rate constant calculated by Lifshitz et al. [18] has a comparable activation energy with that of reaction R1303, which is considered to be too high. The rate constant calculated by Gupte et al. [29] is quantitatively too low compared to those of reactions R1302 and R1303. It is noted that the rate constant calculated by Bentz et al. [17] is a total rate constant, not only for reaction $\text{CH}_3\text{CHO} + \text{H} = \text{CH}_3\text{CO} + \text{H}_2$. Compared to Bentz et al. [17], the result of Sivaramakrishnan et al. [30] has a comparable activation energy while a slightly lower rate. As studied by Bentz et al., H-abstraction from the formyl group is the dominant pathway compared to others. Therefore, the two rate constants from Sivaramakrishnan et al. and Bentz et al. are estimated to have fairly good agreement. In addition, the rate calculated by Sivaramakrishnan et al. is higher than those of reactions R1302 and R1303, but has a lower activation energy, resulting in better simulation results on ignition delay times. Finally, the rate from Sivaramakrishnan et al. was adopted in the improved model. In short, the branching ratio of abstractions from different sites and the effect of the reaction rates on simulation results were both taken into account in the determination of reaction rate constant. Although these methods are relatively unsophisticated, they are useful and effective to some degree before accurate rate constants are obtained through either specific experiments or high-level calculations.

The thermodynamic data of *n*-butanal and the relevant radicals were consistently derived from the C4 model. The detailed *n*-butanal sub-model including 62 reversible reactions is given in Table 2 with references and/or resources of the reactions provided in the last column.

4. Results and discussion

Two selected conditions from the literature were first repeated for comparison using our shock tube facility. A correlation was developed based on all these measured values. The improved *n*-butanal sub-model was then validated against the measured ignition delay times. Reaction pathway and sensitivity analysis were performed on the basis of the new improved *n*-butanal sub-model.

4.1. Experimental results

Two experimental conditions conducted by Davidson et al. [22] were first repeated for comparison, as shown in Fig. 4. It was found

Table 2*n*-Butanal sub-model (units: cm, mol, s, cal, K).

NUM	Reaction	<i>A</i>	<i>n</i>	<i>E_a</i>	Refs.
<i>Unimolecular decomposition</i>					
R1296	NC ₃ H ₇ CHO = NC ₃ H ₇ + HCO	2.84E+22	−1.7	84,460	[28]
R1297	NC ₃ H ₇ CHO = C ₂ H ₅ + CH ₂ CHO	1.78E+22	−1.8	80,670	[28]
R1298	NC ₃ H ₇ CHO = CH ₃ + CH ₂ CH ₂ CHO	6.31E+19	−0.8	87,680	[28]
R1299	NC ₃ H ₇ CHO = NC ₃ H ₇ CO + H	4.96E+16	−0.3	89,070	[7]
R1300	NC ₃ H ₇ CHO = C ₂ H ₅ CHCO + H ₂	3.00E+14	0	84,000	[16]
<i>Hydrogen abstraction</i>					
R1301	NC ₃ H ₇ CHO + H = NC ₃ H ₇ CO + H ₂	1.31E+05	2.6	1220	[30]
R1302	NC ₃ H ₇ CHO + H = C ₃ H ₆ CHO − 1 + H ₂	6.66E+05	2.5	6756	[25]
R1303	NC ₃ H ₇ CHO + H = C ₃ H ₆ CHO − 2 + H ₂	1.30E+06	2.4	4471	[25]
R1304	NC ₃ H ₇ CHO + H = C ₃ H ₆ CHO − 3 + H ₂	2.72E+03	3.1	5210	[30]
R1305	NC ₃ H ₇ CHO + OH = NC ₃ H ₇ CO + H ₂ O	3.37E+12	0	−619	[20]
R1306	NC ₃ H ₇ CHO + OH = C ₃ H ₆ CHO − 1 + H ₂ O	5.28E+09	1	1586	[25]
R1307	NC ₃ H ₇ CHO + OH = C ₃ H ₆ CHO − 2 + H ₂ O	4.67E+07	1.6	−35	[25]
R1308	NC ₃ H ₇ CHO + OH = C ₃ H ₆ CHO − 3 + H ₂ O	1.60E+13	0	2000	[31]
R1309	NC ₃ H ₇ CHO + O = NC ₃ H ₇ CO + OH	5.84E+12	0	1810	[31]
R1310	NC ₃ H ₇ CHO + O = C ₃ H ₆ CHO − 1 + OH	9.81E+05	2.4	4750	[25]
R1311	NC ₃ H ₇ CHO + O = C ₃ H ₆ CHO − 2 + OH	5.49E+05	2.5	3140	[25]
R1312	NC ₃ H ₇ CHO + O = C ₃ H ₆ CHO − 3 + OH	4.77E+04	2.7	2106	[21]
R1313	NC ₃ H ₇ CHO + CH ₃ = NC ₃ H ₇ CO + CH ₄	7.08E−04	4.6	1966	[29]
R1314	NC ₃ H ₇ CHO + CH ₃ = C ₃ H ₆ CHO − 1 + CH ₄	4.53E−01	3.6	7154	[25]
R1315	NC ₃ H ₇ CHO + CH ₃ = C ₃ H ₆ CHO − 2 + CH ₄	6.40E+04	2.2	7520	[25]
R1316	NC ₃ H ₇ CHO + CH ₃ = C ₃ H ₆ CHO − 3 + CH ₄	6.00E+12	0	11,000	[16]
R1317	NC ₃ H ₇ CHO + C ₂ H ₅ = NC ₃ H ₇ CO + C ₂ H ₆	7.08E−04	4.6	1966	– ^a
R1318	NC ₃ H ₇ CHO + C ₂ H ₅ = C ₃ H ₆ CHO − 1 + C ₂ H ₆	1.58E+11	0	12,300	[25]
R1319	NC ₃ H ₇ CHO + C ₂ H ₅ = C ₃ H ₆ CHO − 2 + C ₂ H ₆	1.00E+11	0	10,400	[25]
R1320	NC ₃ H ₇ CHO + C ₂ H ₅ = C ₃ H ₆ CHO − 3 + C ₂ H ₆	6.00E+12	0	11,000	– ^a
R1321	NC ₃ H ₇ CHO + O ₂ = NC ₃ H ₇ CO + HO ₂	3.01E+13	0	39,180	[31]
R1322	NC ₃ H ₇ CHO + O ₂ = C ₃ H ₆ CHO − 1 + HO ₂	3.00E+13	0	52,290	[25]
R1323	NC ₃ H ₇ CHO + O ₂ = C ₃ H ₆ CHO − 2 + HO ₂	2.00E+13	0	49,640	[25]
R1324	NC ₃ H ₇ CHO + O ₂ = C ₃ H ₆ CHO − 3 + HO ₂	2.00E+14	0.5	48,600	[16]
R1325	NC ₃ H ₇ CHO + HO ₂ = NC ₃ H ₇ CO + H ₂ O ₂	3.01E+12	0	11918.7	[28]
R1326	NC ₃ H ₇ CHO + HO ₂ = C ₃ H ₆ CHO − 1 + H ₂ O ₂	4.05E+04	2.5	16,690	[25]
R1327	NC ₃ H ₇ CHO + HO ₂ = C ₃ H ₆ CHO − 2 + H ₂ O ₂	5.88E+04	2.5	14,860	[25]
R1328	NC ₃ H ₇ CHO + HO ₂ = C ₃ H ₆ CHO − 3 + H ₂ O ₂	1.00E+12	0	12,000	[16]
R1329	NC ₃ H ₇ CHO + CH ₃ O = NC ₃ H ₇ CO + CH ₃ OH	1.02E+11	0	2979	[32]
R1330	NC ₃ H ₇ CHO + CH ₃ O = C ₃ H ₆ CHO − 1 + CH ₃ OH	1.50E+11	0	7000	[25]
R1331	NC ₃ H ₇ CHO + CH ₃ O = C ₃ H ₆ CHO − 2 + CH ₃ OH	3.00E+11	0	7000	[25]
R1332	NC ₃ H ₇ CHO + CH ₃ O = C ₃ H ₆ CHO − 3 + CH ₃ OH	5.10E+10	0	2979	[32]
R1333	NC ₃ H ₇ CHO + CH ₃ O ₂ = NC ₃ H ₇ CO + CH ₃ O ₂ H	4.09E+04	2.5	10,200	[25]
R1334	NC ₃ H ₇ CHO + CH ₃ O ₂ = C ₃ H ₆ CHO − 1 + CH ₃ O ₂ H	2.38E+04	2.5	16,490	[25]
R1335	NC ₃ H ₇ CHO + CH ₃ O ₂ = C ₃ H ₆ CHO − 2 + CH ₃ O ₂ H	9.64E+03	2.6	13,910	[25]
R1336	NC ₃ H ₇ CHO + CH ₃ O ₂ = C ₃ H ₆ CHO − 3 + CH ₃ O ₂ H	3.44E+12	0.1	17,880	[25]
<i>Isomerization of first-formed radicals</i>					
R1337	C ₃ H ₆ CHO − 1 = NC ₃ H ₇ CO	2.00E+12	0	36,791	[33]
R1338	C ₃ H ₆ CHO − 2 = NC ₃ H ₇ CO	2.00E+12	0	36,791	[33]
<i>Decomposition of first-formed radicals</i>					
R1339	NC ₃ H ₇ CO = NC ₃ H ₇ + CO	1.07E+12	0.6	16,900	[34]
R1340	NC ₃ H ₇ CO = C ₂ H ₅ CHCO + H	1.00E+13	0	27,960	[35]
R1341	NC ₃ H ₇ CO = C ₂ H ₅ + CH ₂ CO	3.50E+12	0.5	29,470	[36] ^b
R1342	C ₃ H ₆ CHO − 1 = C ₂ H ₄ + CH ₂ CHO	7.40E+12	0	21,970	[25]
R1343	C ₃ H ₆ CHO − 3 = SC ₃ H ₅ CHO + H	4.95E+12	−0.1	31,300	– ^c
R1344	C ₃ H ₆ CHO − 3 = C ₂ H ₅ CHCO + H	8.43E+15	−0.6	40,400	[25]
R1345	C ₃ H ₆ CHO − 3 = C ₂ H ₅ CHO + CH ₃	3.17E+14	−0.4	29,900	[25]
R1346	C ₃ H ₆ CHO − 2 = SC ₃ H ₅ CHO + H	4.95E+12	−0.1	31,300	[25]
R1347	C ₃ H ₆ CHO − 2 = C ₃ H ₆ + HCO	1.26E+13	0	30,340	[25]
R1348	C ₂ H ₅ CHCO + OH = NC ₃ H ₇ + CO ₂	3.73E+12	0	−1010	[25]
R1349	C ₂ H ₅ CHCO + H = NC ₃ H ₇ + CO	4.40E+12	0	1459	[25]
R1350	C ₂ H ₅ CHCO + O = C ₃ H ₆ + CO ₂	3.20E+12	0	−437	[25]
R1351	SC ₃ H ₅ CHO + OH = SC ₃ H ₅ CO + H ₂ O	2.69E+10	0.8	−340	[25]
R1352	SC ₃ H ₅ CO = C ₃ H ₅ − S + CO	8.60E+15	0	23,000	[25]
R1353	SC ₃ H ₅ CHO + HO ₂ = SC ₃ H ₅ CO + H ₂ O ₂	1.00E+12	0	11,920	[25]
R1354	SC ₃ H ₅ CHO + CH ₃ = SC ₃ H ₅ CO + CH ₄	3.98E+12	0	8700	[25]
R1355	SC ₃ H ₅ CHO + O = SC ₃ H ₅ CO + OH	7.18E+12	0	1389	[25]
R1356	SC ₃ H ₅ CHO + O ₂ = SC ₃ H ₅ CO + HO ₂	4.00E+13	0	37,600	[25]
R1357	SC ₃ H ₅ CHO + H = SC ₃ H ₅ CO + H ₂	2.60E+12	0	2600	[25]

^a Estimated based on the reaction NC₃H₇CHO + CH₃.^b Estimated based on the reaction CH₃CH₂CH₂CH₂ = CH₃CH₂ + CH₂CH₂ [36].^c Estimated based on reaction C₃H₆CHO − 2 = SC₃H₅CHO + H.

that the new measured data agree fairly well with theirs, especially at relatively high temperatures. Only shorter ignition delay times of the present study are observed in the relatively low-tempera-

ture range at *p* = 1.4 atm. The measured ignition delay times at 2.6 atm agree well with those of Davison et al. [22] across the entire temperature range.

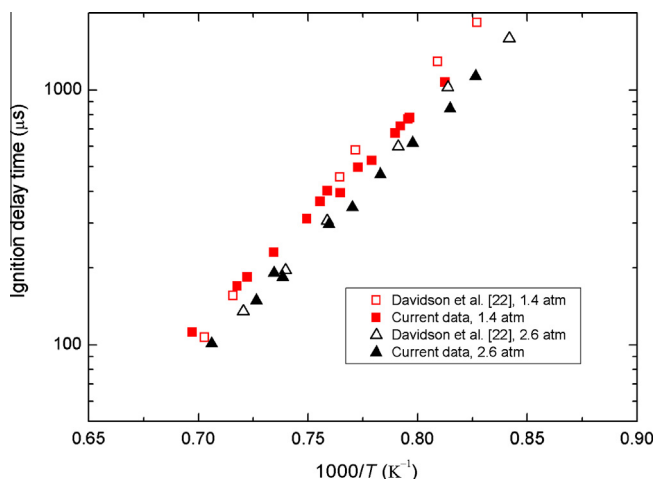


Fig. 4. Comparison with previous data conducted by Davidson et al. [22] at $\phi = 1.0$, $p = 1.4$ atm and 2.6 atm for 1% *n*-butanal.

Measurements were then performed under various temperatures, T , pressures, p , and equivalence ratios, ϕ . A correlation of the ignition delay time using a previous formula suggested by

Horning et al. [37] is obtained with multiple linear regression method based on all the experimental data.

$$\tau = 4.66 \times 10^{-15} \times [n\text{-butanal}]^{0.37 \pm 0.10} \times [\text{O}_2]^{-1.06 \pm 0.03} \times [\text{Ar}]^{0.31 \pm 0.07} \times \exp(39.7 \pm 0.6/RT) \quad (1)$$

where τ is the ignition delay time in seconds, pressure is in atmospheres, concentrations are in moles per cubic centimeter, τ is the temperature in Kelvin, activation energy is in kilocalories per mole, and $R = 1.986 \times 10^{-3}$ kcal/(mole K) is the universal gas constant. The regression coefficient $R^2 = 0.978$ indicates a fairly satisfactory regression. Extrapolating this correlation to other conditions beyond this study should be conducted with caution.

Figure 5 shows the measured ignition delay times of *n*-butanal mixtures (0.6% and 1.2%) at three pressures ($p = 1.3, 5$ and 10 atm) for the fuel-lean ($\phi = 0.5$), stoichiometric ($\phi = 1.0$), and fuel-rich mixtures ($\phi = 2.0$). All the measured ignition delay times and the improved mechanism are provided as [Supplementary data](#). In general, the measured ignition delay times exhibit good Arrhenius exponential dependence on the reciprocal temperature, and increase with increasing equivalence ratio at three pressures. The correlations well reproduce the measured ignition delay times both qualitatively and quantitatively.

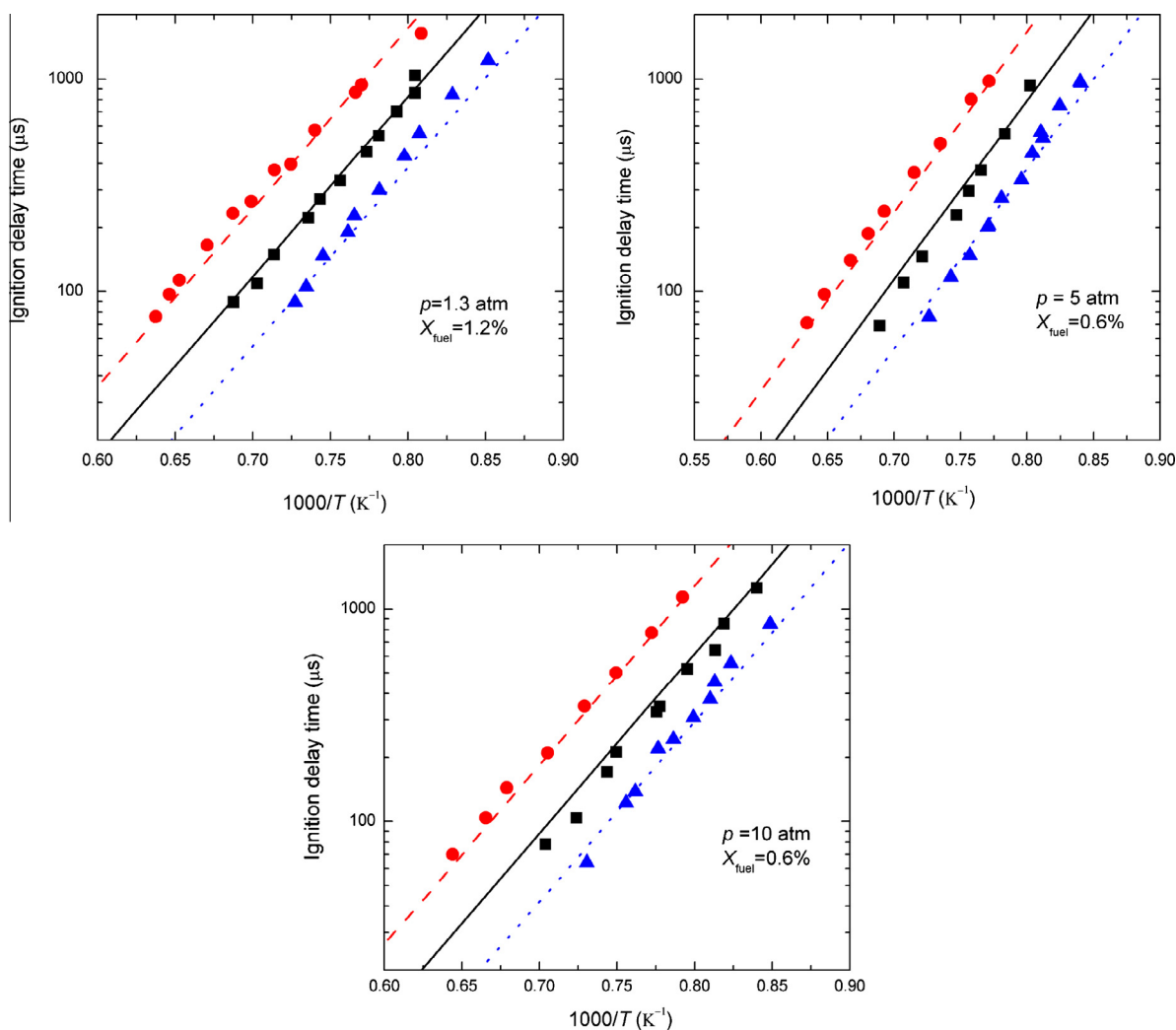


Fig. 5. Measured ignition delay times of *n*-butanal under various conditions (Symbols (experiment): $\phi = 0.5$ ($\blacktriangle$), $\phi = 1.0$ ($\blacksquare$), $\phi = 2.0$ ($\bullet$); Lines (correlation): $\phi = 0.5$ (blue dot line), $\phi = 1.0$ (black solid line), $\phi = 2.0$ (red dash line)). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

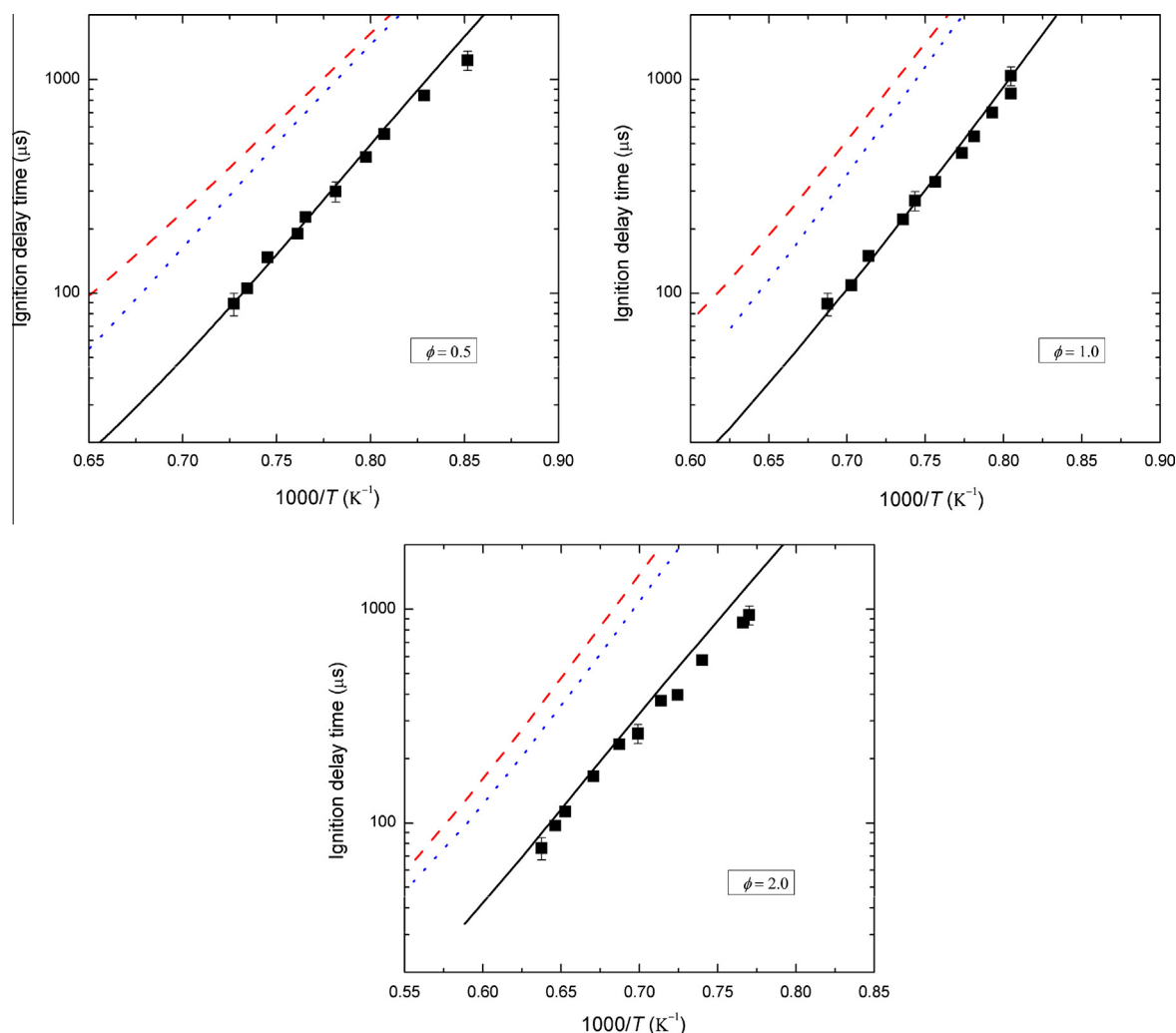


Fig. 6. Comparison between experimental and modeling results at $p = 1.3$ atm (symbols: experimental data; dash line: C4 model; dot line: veloo model; solid line: improved model).

4.2. Simulation results and validation against the measured data

Simulation of ignition delay times was performed using Chemkin II [24] in a constant-volume, adiabatic and zero-dimensional reactor. Two previously published *n*-butanal models, a C4 model [25] and the Veloo model [21], were also employed for comparison. It is noted that the C4 model primarily aims at C1–C4 alkanes, with only a tentative *n*-butanal model included. It lacks some important unimolecular decomposition and H-abstraction reactions and was not validated. Veloo et al. [21] validated the propanal sub-model but did not validate the *n*-butanal sub-model in their study. A recent *n*-butanol model developed by Harper et al. [8] also includes the *n*-butanal chemistry, which was validated against the ignition data by Davidson et al. [22] in their study. Nevertheless, this model was found to be very computationally inefficient even for the zero-dimensional ignition calculation and was not employed in this study.

Figures 6–8 show the comparison between experimental and modeling results under various conditions. The proposed *n*-butanal model agrees well with the experimental results at both various pressures and equivalence ratios, while the C4 and Veloo models vastly over-predict the measured ignition delay times. At $p = 1.3$ atm, the Veloo model predicts well the activation energy while the C4 model slightly under-predicts it. It is of interest to

note that the slope exhibits a gradual curvature with decreasing temperature at 10 atm, particularly for the fuel-lean mixture ($\phi = 0.5$). The reason is that the temperature enters the intermediate temperature regime where the hydroperoxyl chemistry becomes more important.

4.3. Reaction pathway and sensitivity analysis

Analysis of reaction pathway was performed for a 1.2% *n*-butanal mixture at $p = 1.3$ atm, $T = 1300$ K and $\phi = 1.0$, as shown in Fig. 9. The timing of 20% fuel consumption is chosen for the analysis as that in the literatures [7,10].

Under this condition, the unimolecular decomposition and H-abstraction reactions contribute to the fuel consumption by about 22% and 78%, respectively. It is observed that the decomposition of the $\text{C}_\alpha\text{--C}_\beta$ bond has the largest branching ratio (10.1%), which is consistent with the calculation by da Silva and Bozzelli [26] that the $\text{C}_\alpha\text{--C}_\beta$ bond has the weakest BDE (82.5 kcal/mol). The decomposition of the $\text{C}_\alpha\text{--CHO}$ bond comes to the second and also has a relatively large branching ratio (8.5%). In addition, sensitivity analysis in the next section shows that these two dissociation reactions exert large influence on the ignition delay. For H-abstraction reactions, H-abstraction from the formyl group is the dominant channel with a branching ratio up to 24.8%. However, the branching

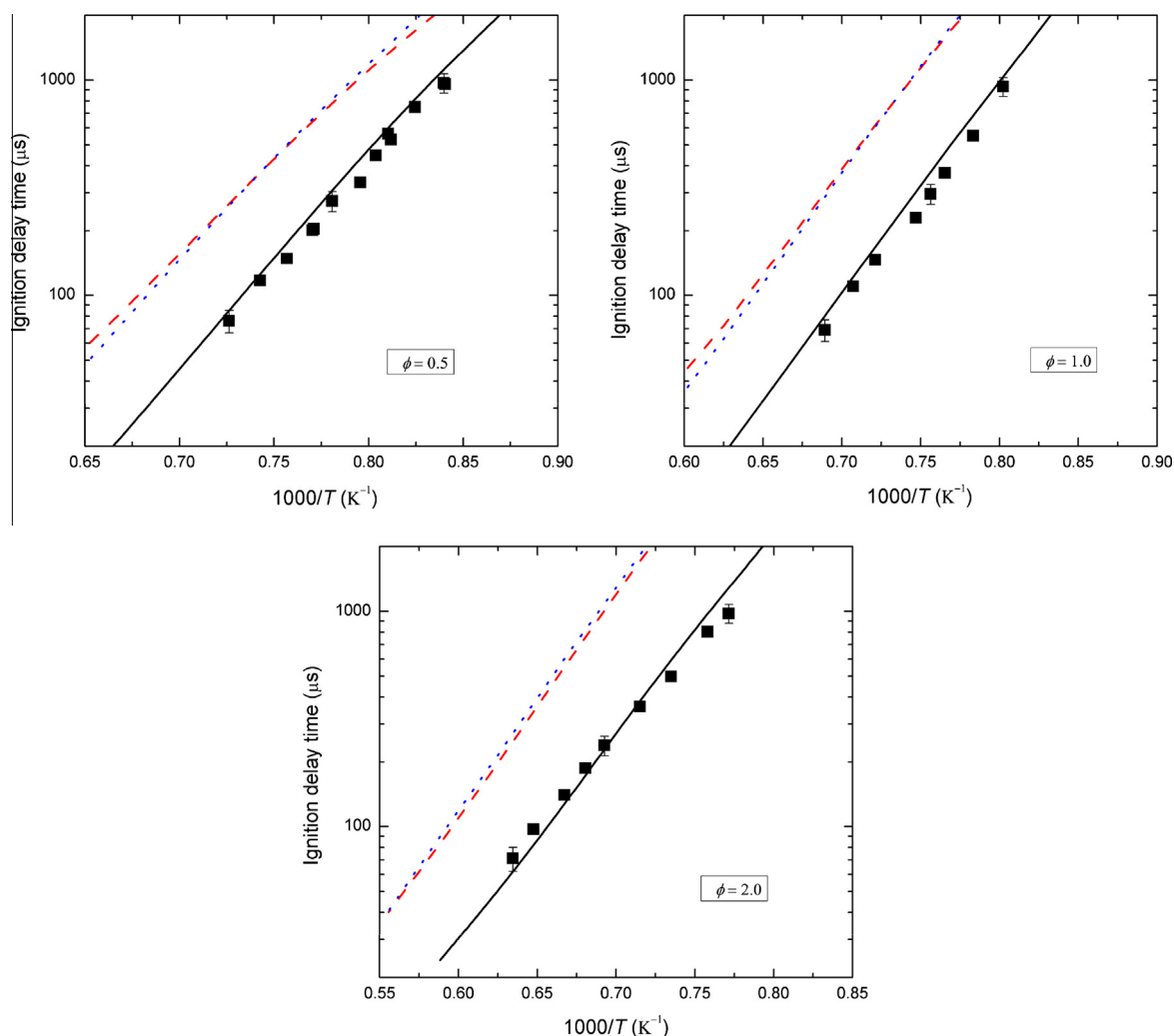


Fig. 7. Comparison between experimental and modeling results at $p = 5.0$ atm (symbols: experimental data; dash line: C4 model; dot line: veloo model; solid line: improved model).

ratio of H-abstraction from the α -carbon site (16.9%) is even smaller than that from β -carbon site (19.7%), which may be too small according to the BDEs. In fact, the rate constants for α -carbon site were obtained based on a tentative analogy with that of H-abstraction from the primary carbon of acetaldehyde. Therefore, more accurate experiment or calculation on these reaction rate constants will further improve the new model. Furthermore, it is found that H-abstraction reactions by H and OH radicals are of remarkable importance. The first radicals formed after H-abstraction mainly undergo β -scission to form smaller radicals at high temperatures. Decomposition of butyryl radical ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}$) almost exclusively decomposes to form propyl radical and carbon monoxide. Similar to the analysis of Veloo et al. [21] for propanal, this channel should also be the dominant pathway in the oxidation of n -butanal at low temperatures, and the formed propyl radicals then undergo the low-temperature branching and lead to the appearance of a cool flame. To better understand the low-temperature oxidation of n -butanal, further experimental and chemical investigations are needed.

4.4. Sensitivity analysis

Sensitivity analysis, conducted by perturbing each reaction rate constant by a factor of 2 was performed for the ignition delay time

of 1.2% n -butanal under the same condition as that for the reaction pathway analysis. Detailed definition was given in the literature [10]. Negative coefficient indicates a promoting effect on the overall reactivity and vice versa. The twenty most sensitive reactions are shown in Fig. 10. It is found that the sensitivity is dominated by the reactions related to small radicals. However, some fuel-specific reactions such as the unimolecular decomposition and H-abstraction reactions also exhibit high sensitivity. It is interesting to note that all three unimolecular decomposition reactions presented in the figure show negative coefficients and exhibit positive influence on the ignition delays, while both two H-abstraction reactions exhibit negative influence. This is because these unimolecular decomposition reactions, being chain-branching reactions, consume fuel molecules and form more active radicals, while the H-abstraction reactions, being chain propagation reactions, consume H radicals and form less active radicals. The large sensitivities of these fuel-specific reactions imply that it is meaningful to carry out experimental or theoretical studies on these reaction rate constants to further optimize the model.

5. Conclusions

Ignition delay times of n -butanal were measured over a wide range of equivalence ratios, pressures and temperatures. An

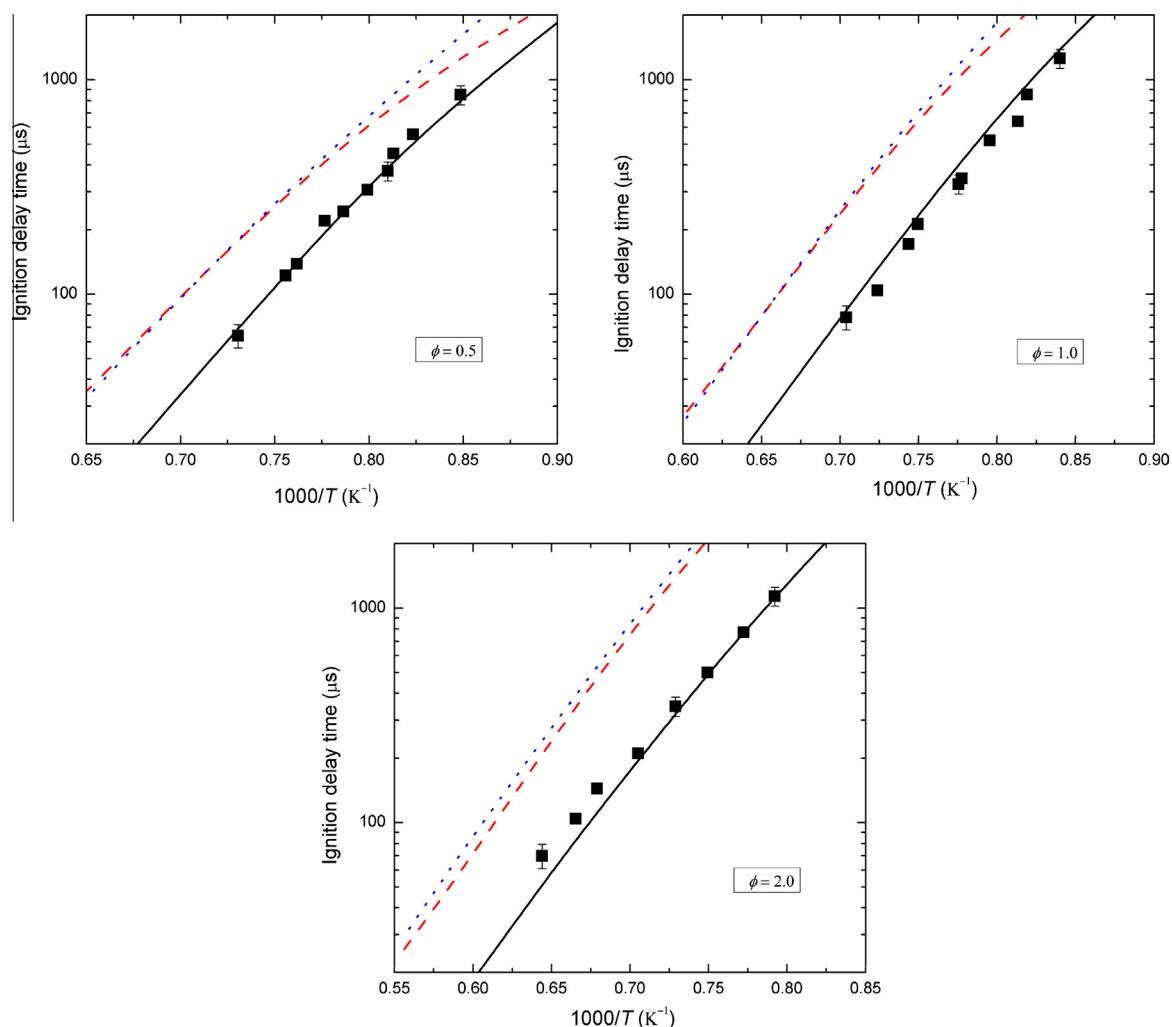


Fig. 8. Comparison between experimental and modeling results at $p = 10.0$ atm (symbols: experimental data; dash line: C4 model; dot line: veloo model; solid line: improved model).

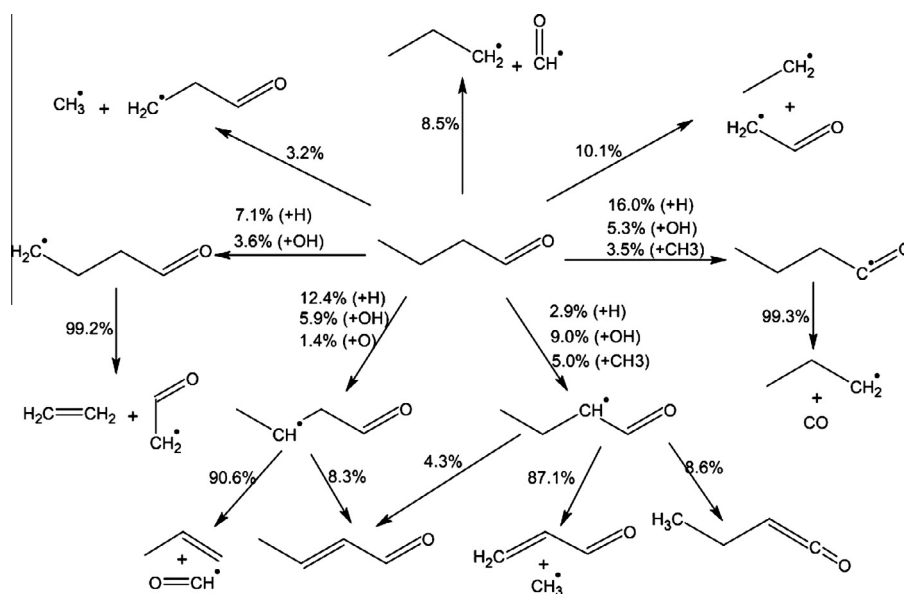


Fig. 9. Reaction pathway diagram of the improved model for 1.2% *n*-butanal in shock tube at $p = 1.3$ atm, $T = 1300$ K, $\phi = 1.0$ and 20% fuel consumption.

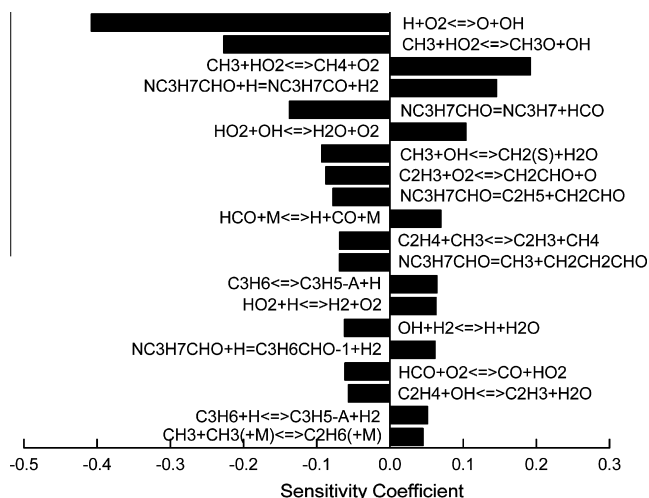


Fig. 10. Sensitivity analysis for 1.2% *n*-butanol at $T = 1300$ K, $p = 1.3$ atm and $\phi = 1.0$.

ignition delay time correlation is obtained based on the measured data using the multiple linear regression method.

A detailed *n*-butanol sub-model was developed on the basis of literature review. Compared to the previous *n*-butanol models, the improved model shows substantial improvements on the prediction of the ignition delays under all measured conditions. Analysis on reaction pathway shows that the breakages at C_{α} -CHO and C_{α} - C_{β} bonds have relatively large branching ratios among the unimolecular decomposition reactions, and H-abstraction from the formyl group is the dominant pathway. Distribution of these reaction pathways is roughly consistent with the magnitude of the BDEs. Finally, sensitivity analysis was performed under a selected condition. To optimize the *n*-butanol sub-model, further validation against other experimental targets, such as JSR or flames, is recommended.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2013.04.002>.

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